面內磁場下1T相二碲化鉑約瑟夫森接面之電流－相位關係研究

Study of the Current – Phase Relation of a 1T-$PtTe_2$ Josephson Junction Under an In-Plane Magnetic Field

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Outline

Introduction

- Research motivation
- 1T-$PtTe_2$ (Type-II Dirac)

Theory

- Josephson effect & CPR
- Josephson diode effect

Methods

- Device fabrication
- Measurement setup

Results

- Basic characterization
- CPR analysis
- In-plane field effect

Research Motivation

Topological SC

- Type-II Dirac semimetal **$PtTe_2$** proximitized by a superconductor

CPR

- Direct probe of junction physics beyond simple $\sin\phi$

Diode Effect

- Non-reciprocal transport $I_c^+ \neq |I_c^-|$
- Route to dissipationless

Theoretical Background

Josephson Effect & Current – Phase Relation

Josephson relations

- DC: $I_s = I_c \sin\phi$ (ideal tunnel junction)
- AC: $\dot{\phi} = \frac{2e}{\hbar} V$

Beyond sinusoidal CPR

- High transparency $\rightarrow$ **skewed** CPR
- Harmonics: $I(\phi) = \sum_n I_n \sin(n\phi + \delta_n)$
- Access via **asymmetric SQUID** readout

Josephson Diode Effect (Concept)

Definition

- Non - reciprocal critical current $I_c^+ \neq |I_c^-|$
- Superconducting rectification without Joule heating

Typical ingredients

- Broken inversion symmetry + broken time - reversal symmetry
- Often tuned by $B_{\|}$ with strong spin – orbit coupling

Theoretical Model: Rashba + Zeeman

Rashba + Zeeman: Why We Care

- Platform: 2D electron system at an interface (e.g. PtTe_2 surface / $\mathrm{S}-\mathrm{PtTe}_2$ junction)
- Key ingredients:
  - Strong spin – orbit coupling (SOC) $\rightarrow$ Rashba effect
  - External magnetic field $\rightarrow$ Zeeman effect
- Combined effect:
  - Generates **helical spin – momentum locking + spin splitting**
  - Breaks **inversion** and **time-reversal** symmetries $\rightarrow$ allows **finite-momentum pairing** and **Josephson diode effect (JDE)**

Minimal 2D Model (Rashba + Zeeman)

- Momentum in plane: $\mathbf{k} = (k_x, k_y)$
- Spin Pauli matrices: $\sigma = (\sigma_x, \sigma_y, \sigma_z)$
- Total Hamiltonian:

Experimental Methods

Device Fabrication Flow

1. **Crystal Growth:** Self-flux method (provided by Prof. C.S. Lue, NCKU).
2. **Exfoliation:** Mechanical exfoliation of $PtTe_2$ flakes onto SiO_2/Si substrates.
3. **E-beam Lithography (EBL):**
   - PMMA resist coating.
   - Pattern definition for contacts and SQUID loops.
4. **Metallization:**
   - Superconducting electrodes: **Al (Aluminum)** or **Nb (Niobium)**.
   - In-situ Ar ion milling to ensure transparent interfaces.
   - E-beam evaporation & Liftoff.
5. **Design:** Asymmetric dc-SQUID geometry for CPR retrieval.

Measurement Setup

- **Cryogenic system:** Dilution refrigerator (base ~40 mK)

Results: Basic Characterization

I-V Characteristics (DC Transport)

- **Typical IV Curve:**
  - Clear Superconducting branch ($V = 0$).
  - Switching Current (I_c) and Retrapping Current (I_r).
  - Hysteretic behavior (underdamped junction).

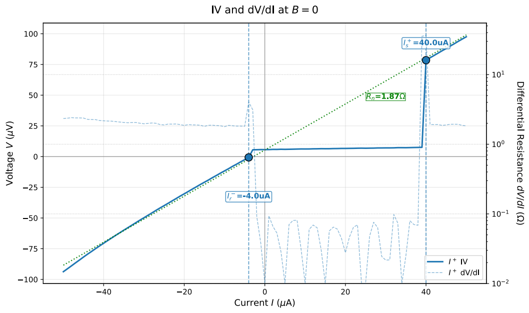


Figure 3: I-V Curve at Base Temp.

- **Observation:**
  - Sharp transition indicates high-quality junction.
  - Finite resistance in normal state (R_N).
  - **Excess Current** observed at high bias $\rightarrow$ High transparency interface.

Results: Current-Phase Relation

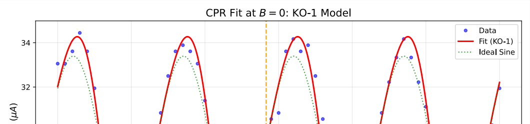
CPR Retrieval Method

- **Asymmetric SQUID Technique:**
  - Large junction (I_{ref}) // Small junction (I_{test}).
  - $I_c(\Phi_{ext}) \approx I_{ref} + I_{test}(\phi)$.
  - Assumes $I_{ref} \gg I_{test}$ and sinusoidal $I_{ref}(\phi)$.
- **Math Model:**

$$I_s(\phi) = I_1 \sin(\phi) + I_2 \sin(2\phi + \delta)$$

  - I_1: Fundamental component (conventional).
  - I_2: Second harmonic (4π-periodic / ABS).

Measured CPR & Fitting

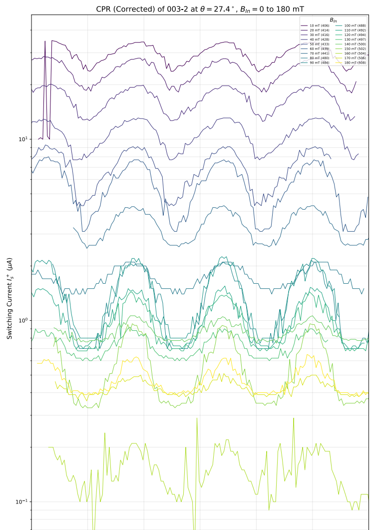


- **Key Findings:**
  - **Skewed CPR:** Deviation from pure sine wave.
  - **High I / I ratio:** Significant second

Results: In-Plane Magnetic Field Effect

Tunable CPR with In-Plane Field

- Applying $B_{\|}$ **modifies CPR shape**
- Suggests a **phase shift** (ϕ_0) in higher harmonics
- Correlates with emergence of the **Josephson diode effect**



Conclusion

Summary

1. **Successful Fabrication:**
   - High-quality 1T-$PtTe_2$ Josephson junctions and SQUIDs realized.
2. **Anomalous CPR:**
   - Observed significant **second harmonic** (I_2) contribution.
   - Direct evidence of high-transparency transport modes.
3. **Magnetic Tuning:**
   - In-plane magnetic field effectively modulates the CPR and induces **Josephson Diode Effect.**
   - Demonstrates potential for **tunable superconducting logic** and **phase batteries.**

Future Outlook

- **Optimization:** Improving fabrication yield and interface control.
- **Material Exploration:** Extending to other Weyl/Dirac semimetals.
- **Quantum Information:** Integration into superconducting qubit circuits (e.g. as

References